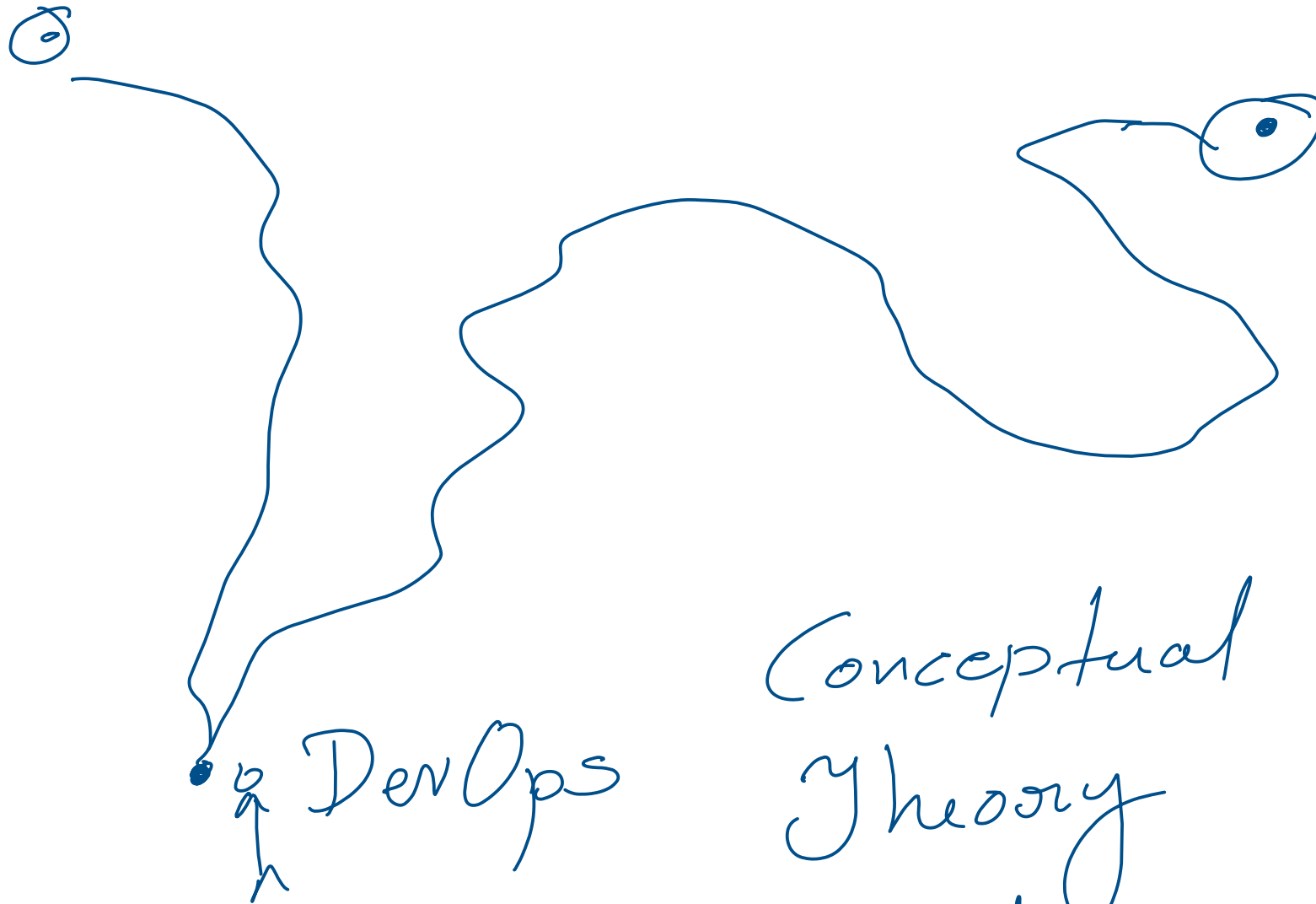


Whiteboard

DevOps 12-Jun-2021 Batch



Conceptual
Theory
Practice

No App Development

- lots of learning -
 - Overwhelm you - concepts
 - Demo's
- DevOps process - new

App Development

— DevOps - tools - processes

QA test Cases

✓ P →

✓ F

App

Web Page(s) ⇒ test

⇒

Production

- jar
- war } Zip

50 fn

1000 concurrent

50 fn

1000 concurrent

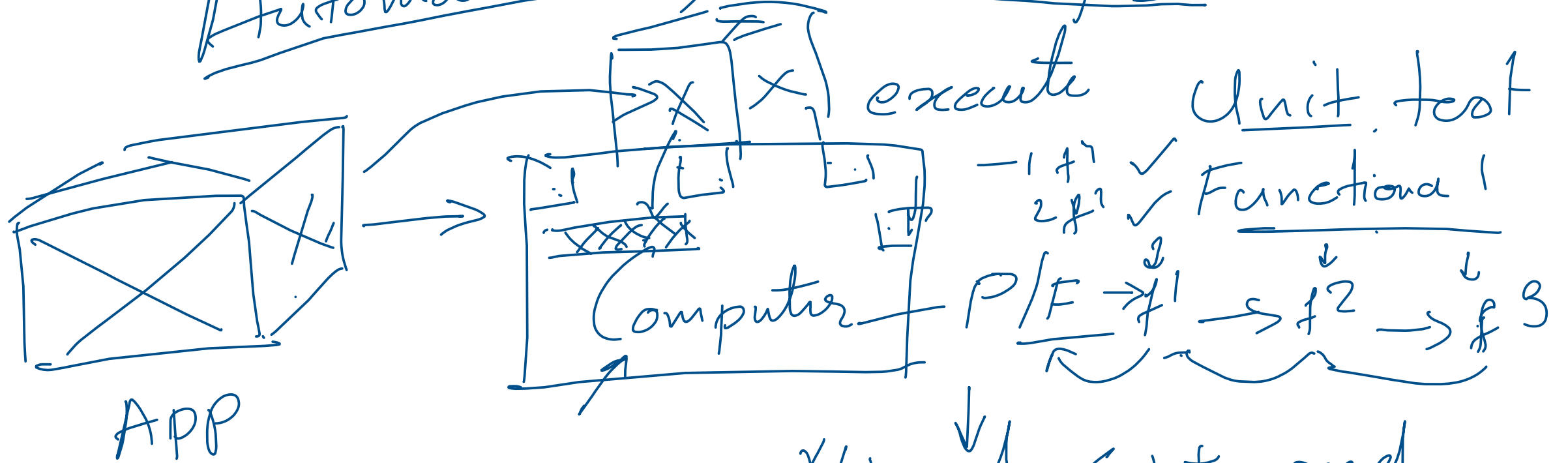
? 2000

500 MB

50 fn

1000
Concurrent

Automated by DevOps

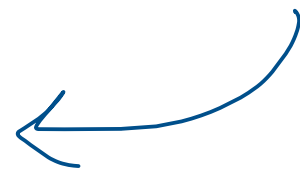


→ MS office

→ Chrome

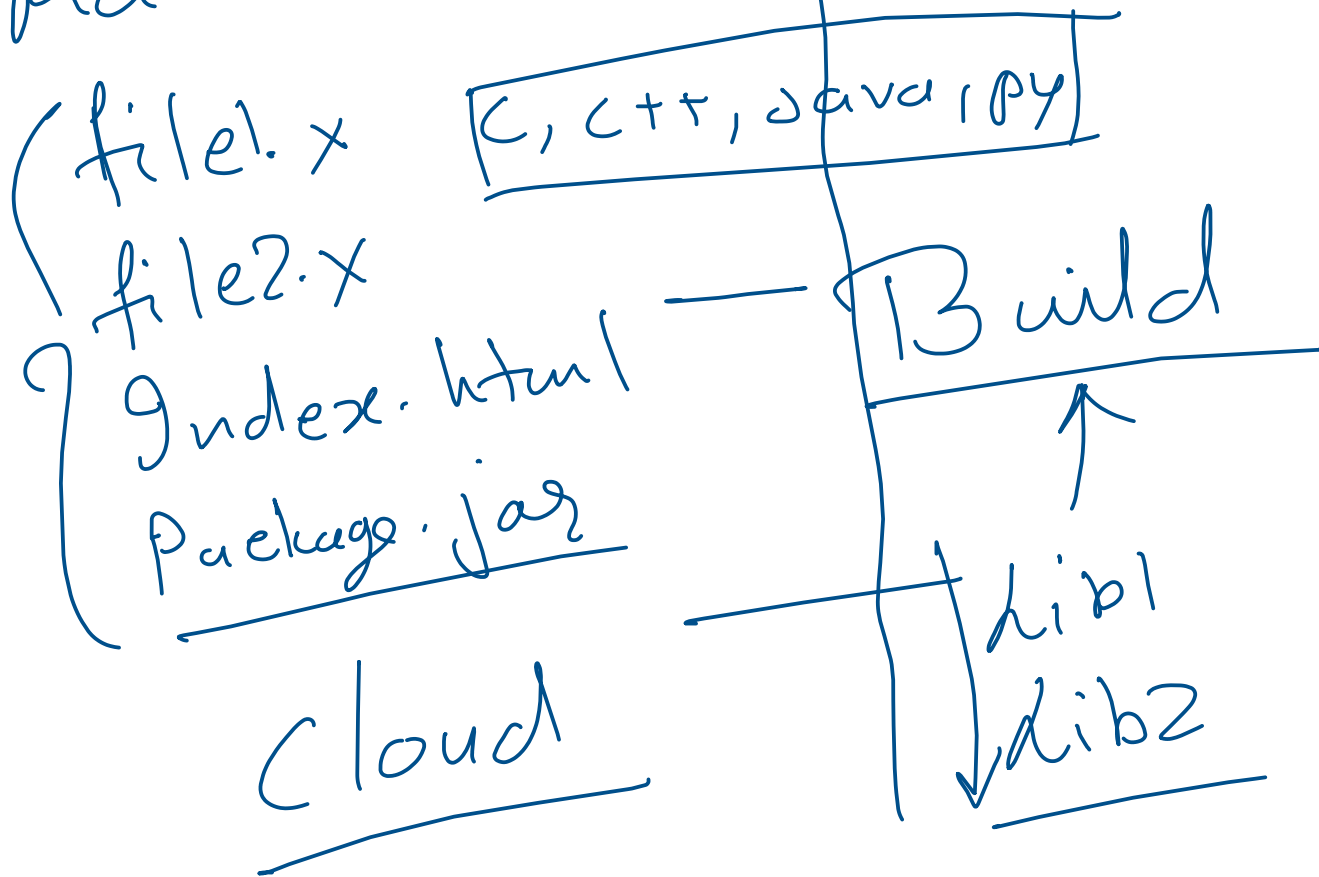
Visual C++ 2015 2019

✓ Compliance



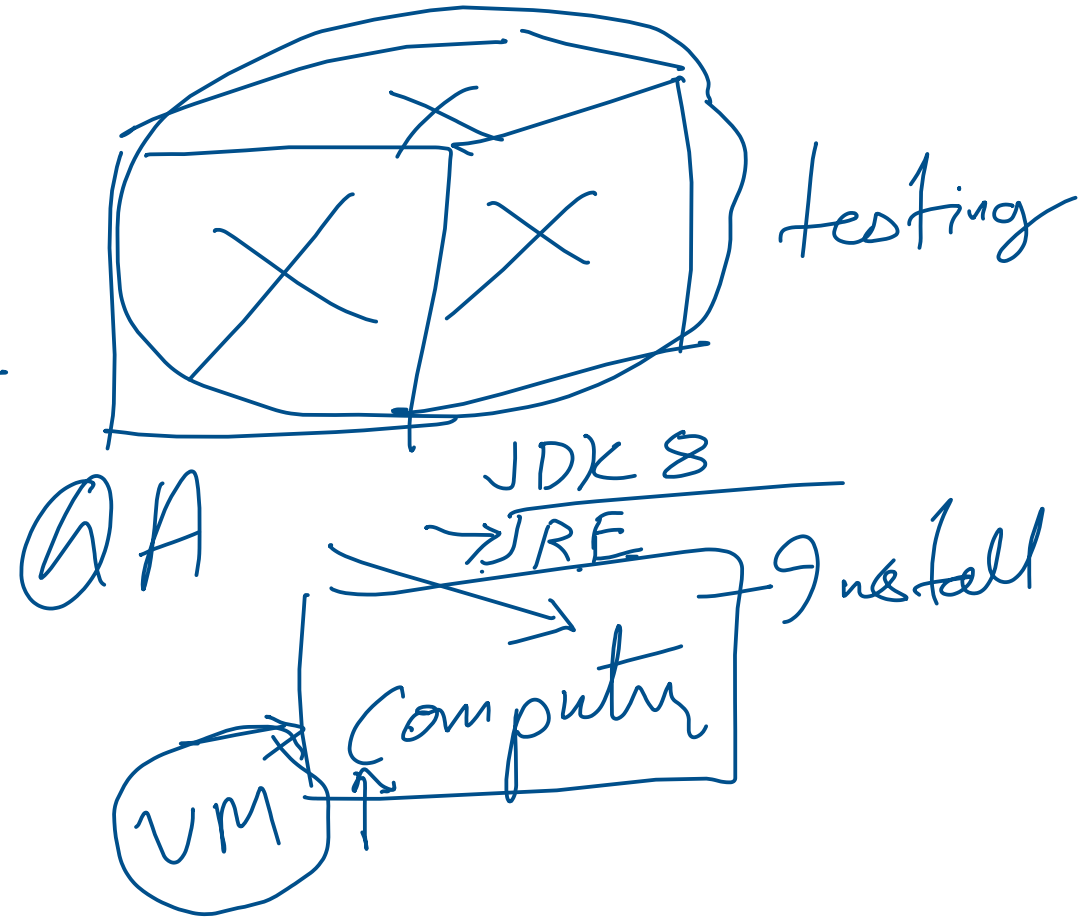
App Package

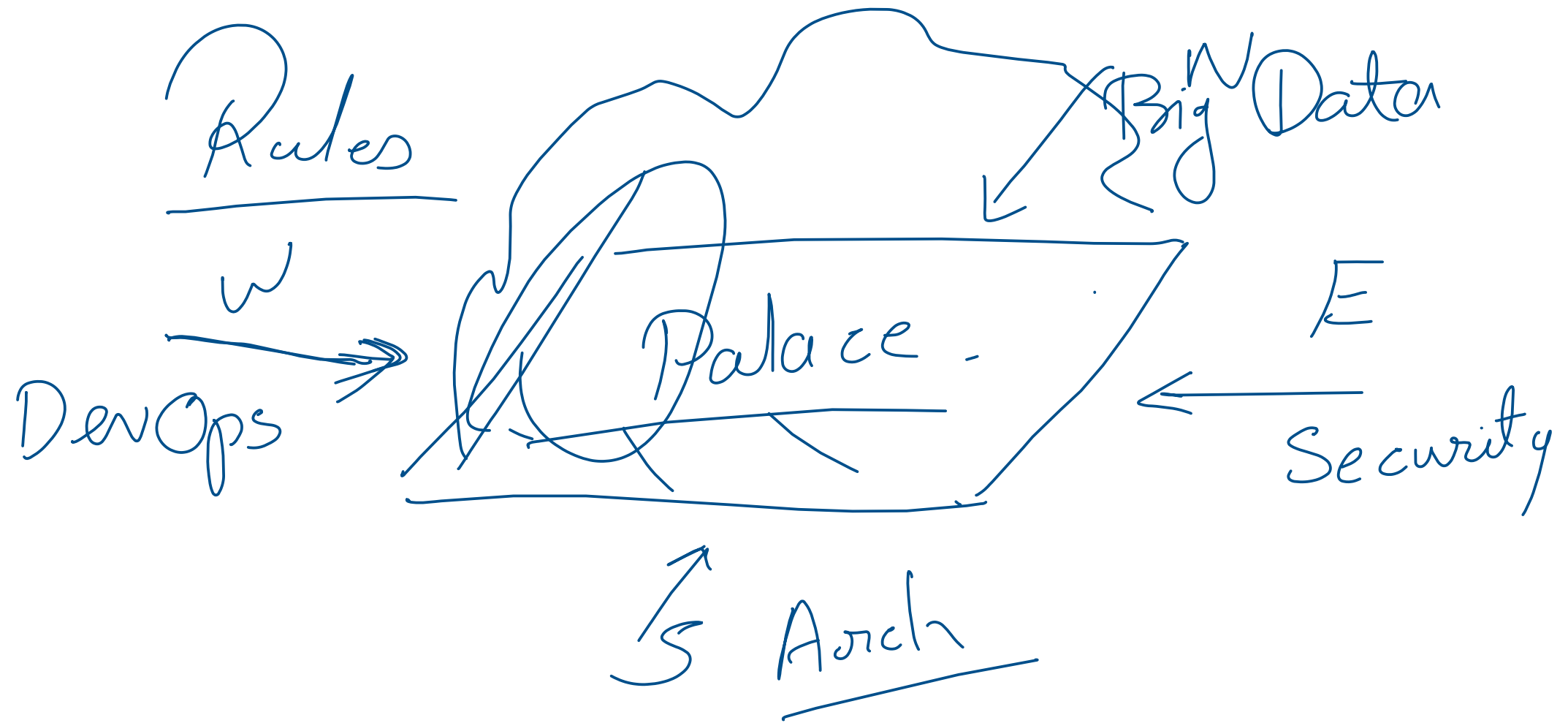
Raw Source Code

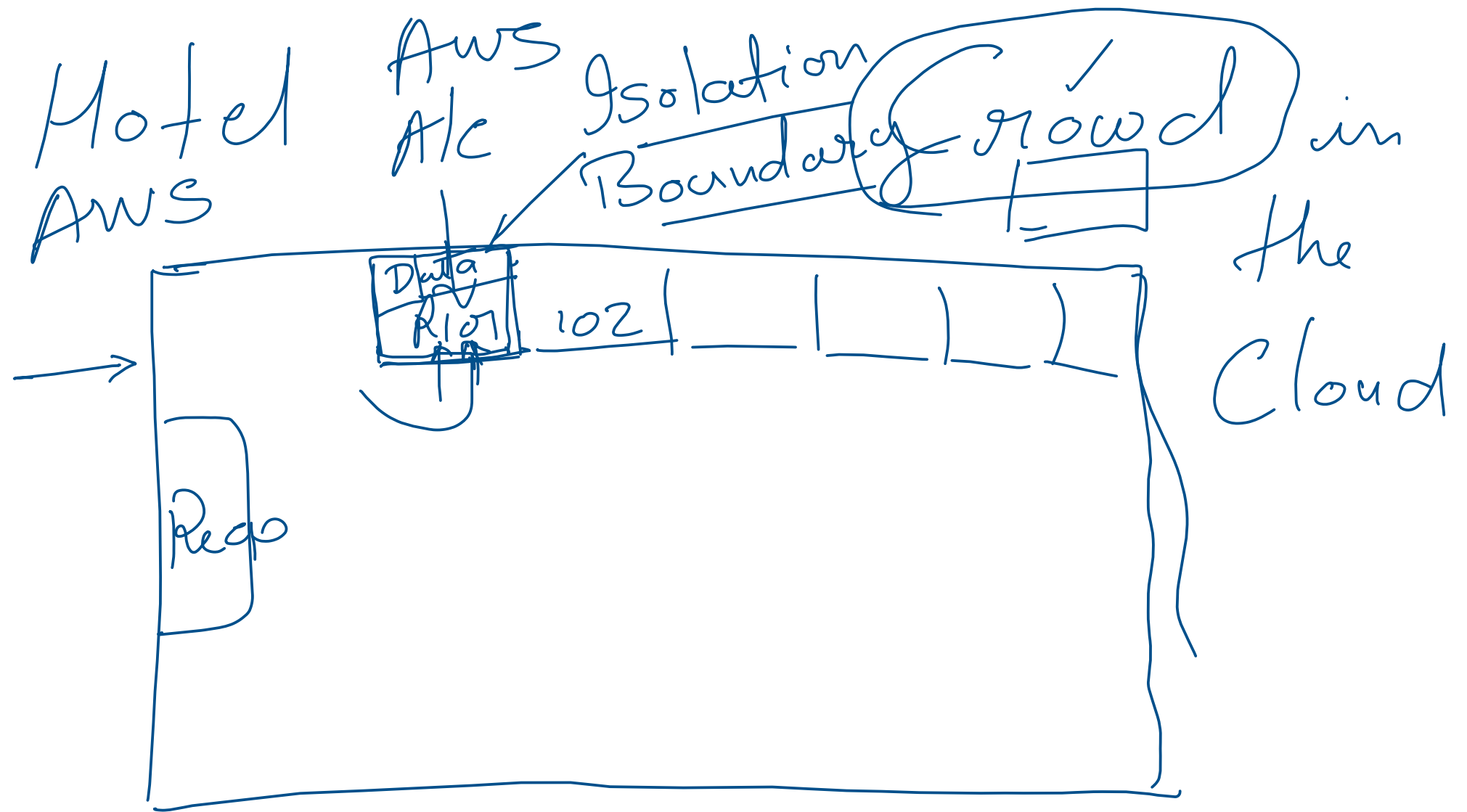


Software

Java — .class





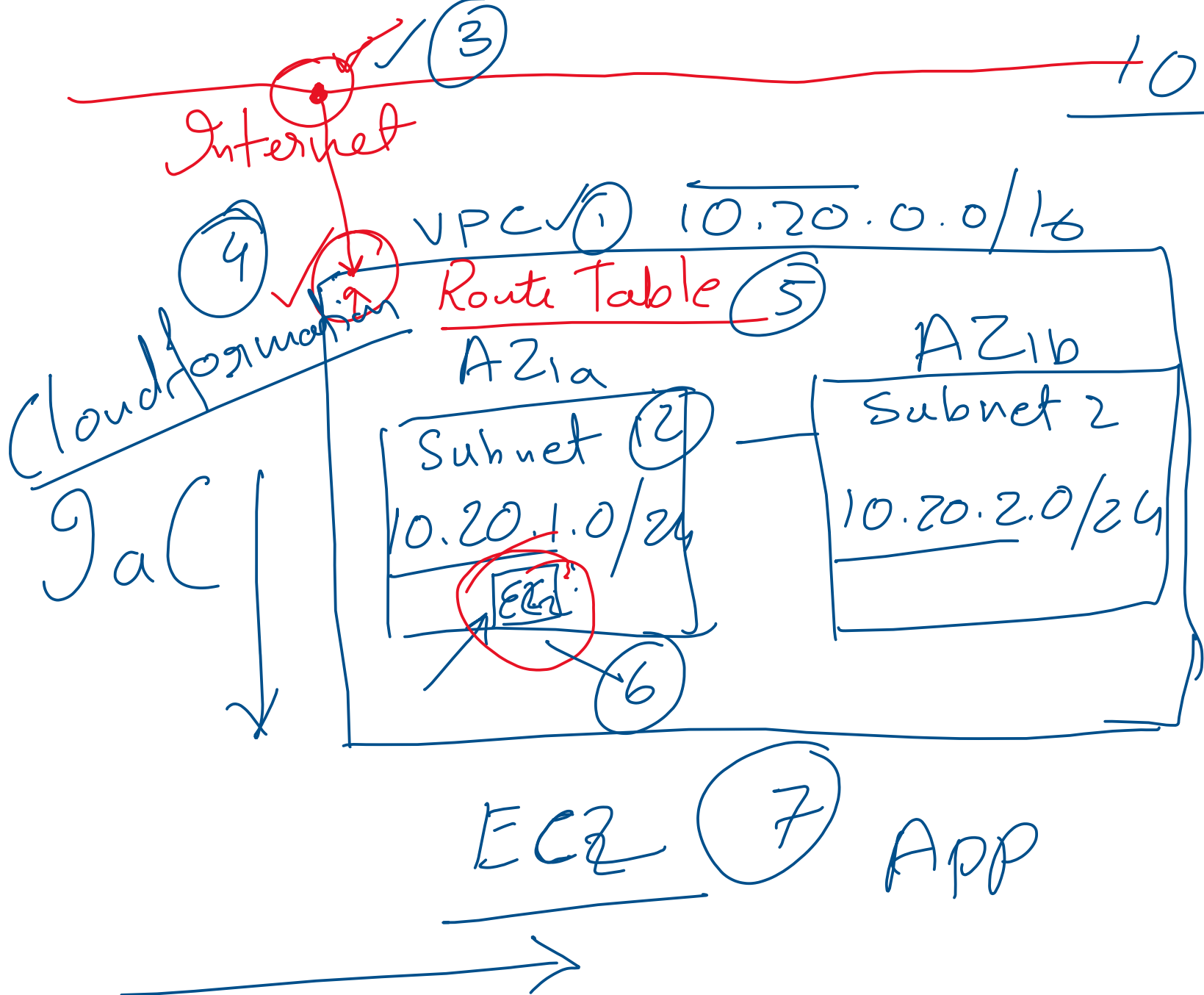


↓
DevSecOps

Service

why, what	defn
How	2 Implementation(s)
How, where	3 Integration
why, what	4 Security
who	5 Error name

at hr 30m



$$\frac{10.10.0.0}{20.0.0} / 16$$

- 1. 255
- 2. 255
- 3. 255

byte

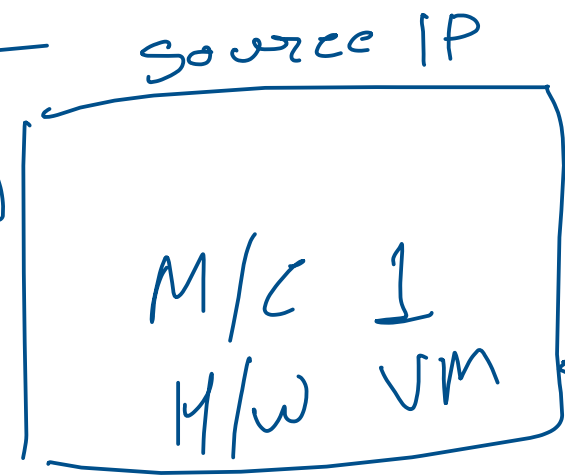
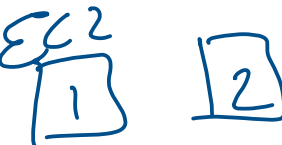
10	10	10	10
----	----	----	----

8 bits

/24

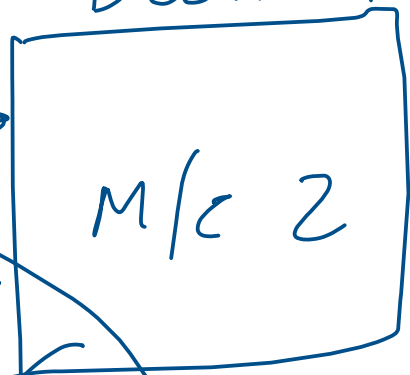
Internet

Source IP Packet Destination IP



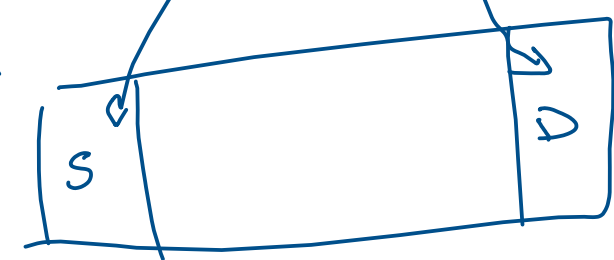
Route Table

Destination IP



Router on-prem

Cloud

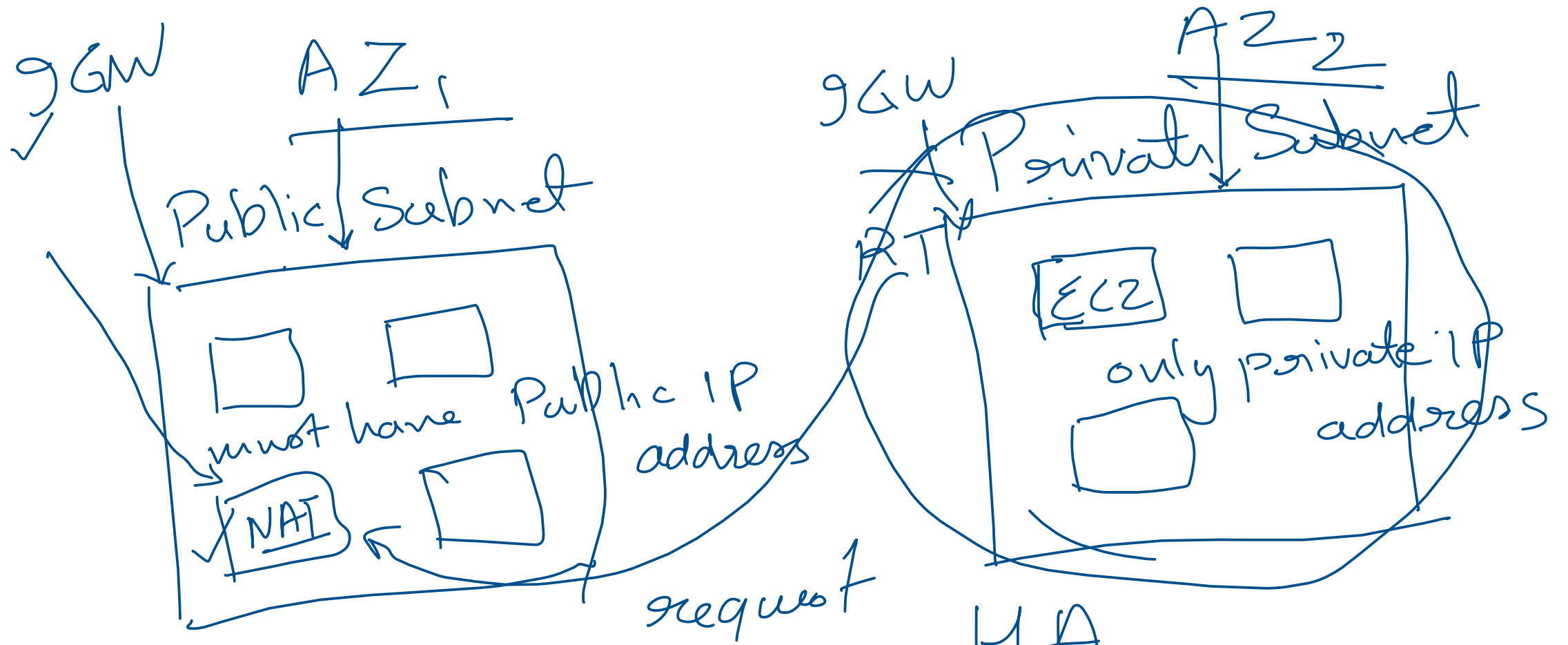


local N/w

gw

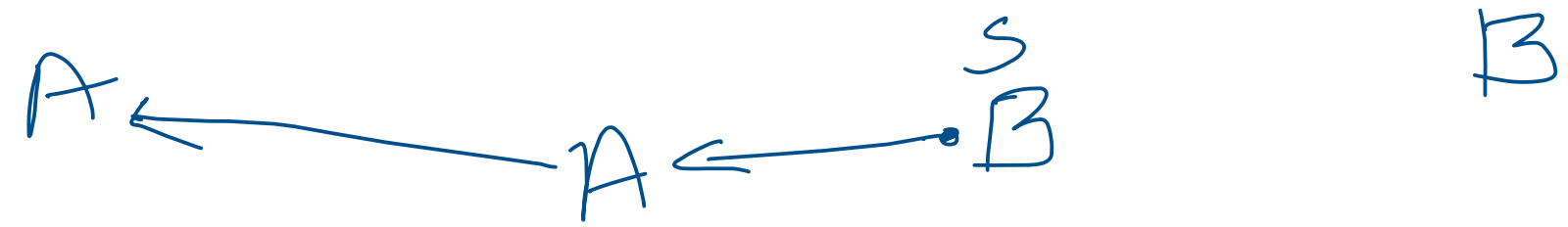
10.20.0.0

0.0.0.0/0



request
A → B
S

HA



EC2 instance

Private IP

→ fixed as per VPC CIDR

Public IP

→ issued from pool

Stop

— IP released
back to
pool

IPv4 → of AWS owned
IP addresses
limited

10,000

Start

— a new
IP from pool

✓ Elastic IP

IP fixed (from pool)

DNS

Public IP

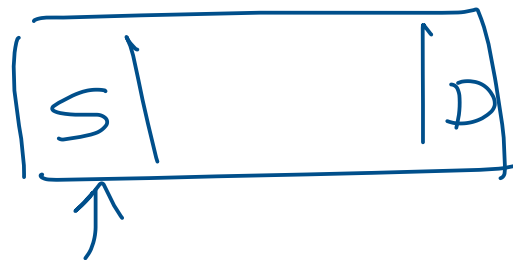
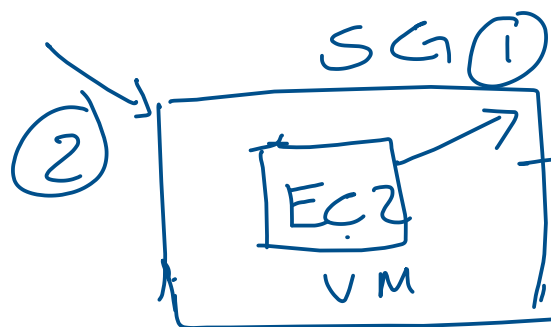
↖ NAT

VM

NACL

Subnet

✓ allow
Deny

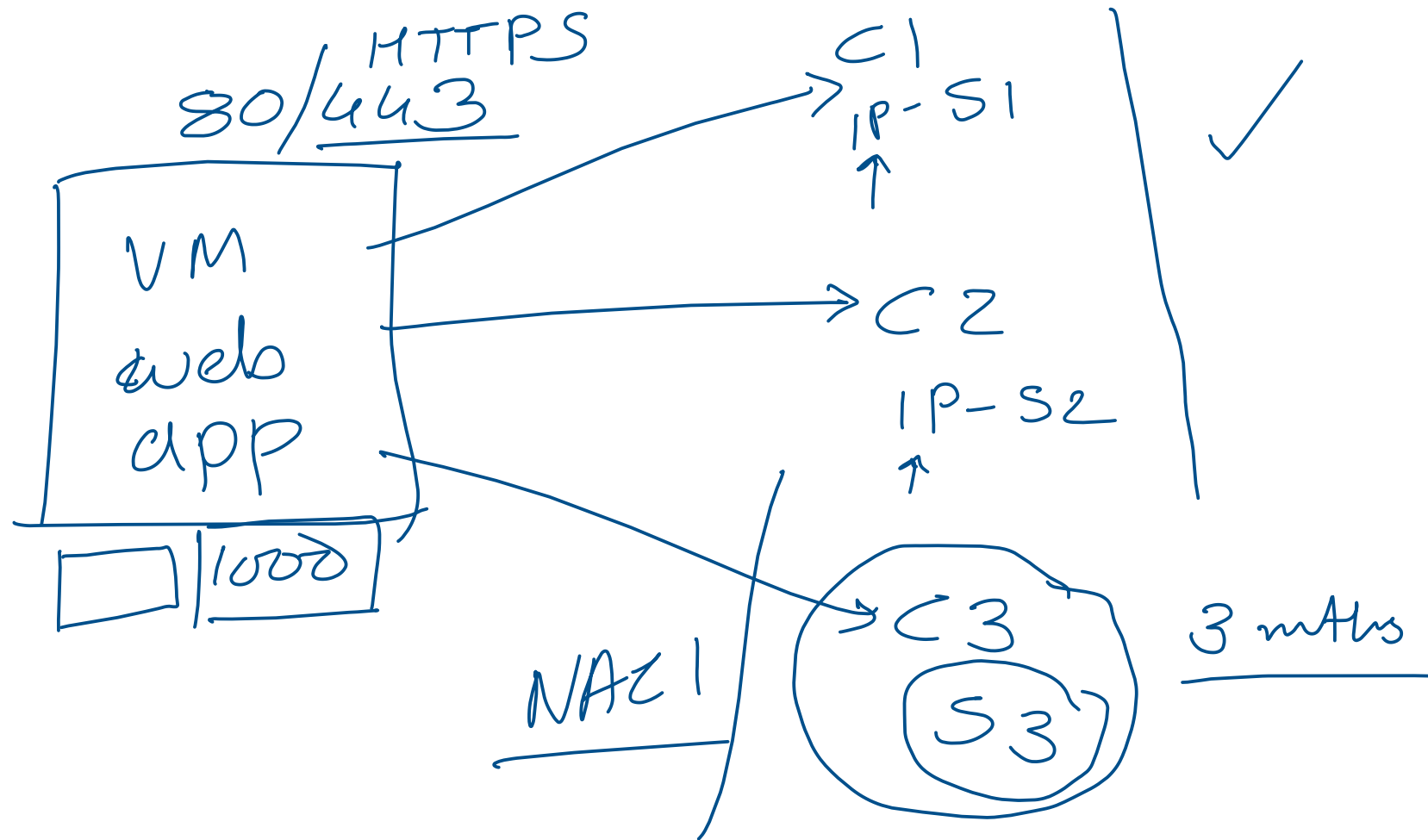


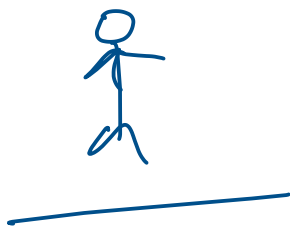
protocol → TCP

Port → 80

Source IP → allow ✓
~~deny~~

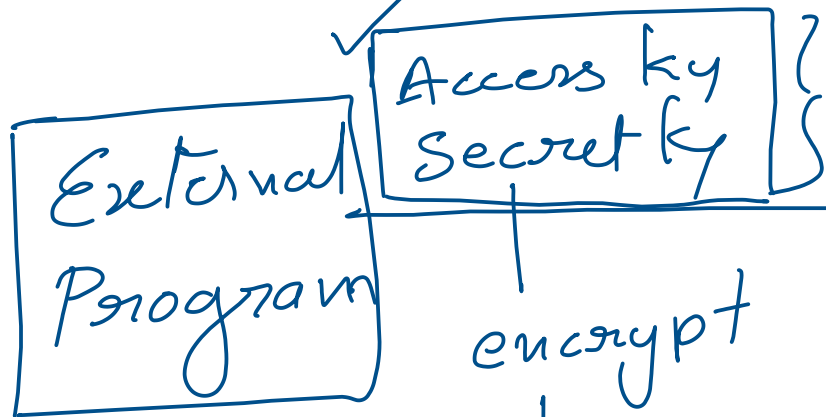
0.0.0.0/0





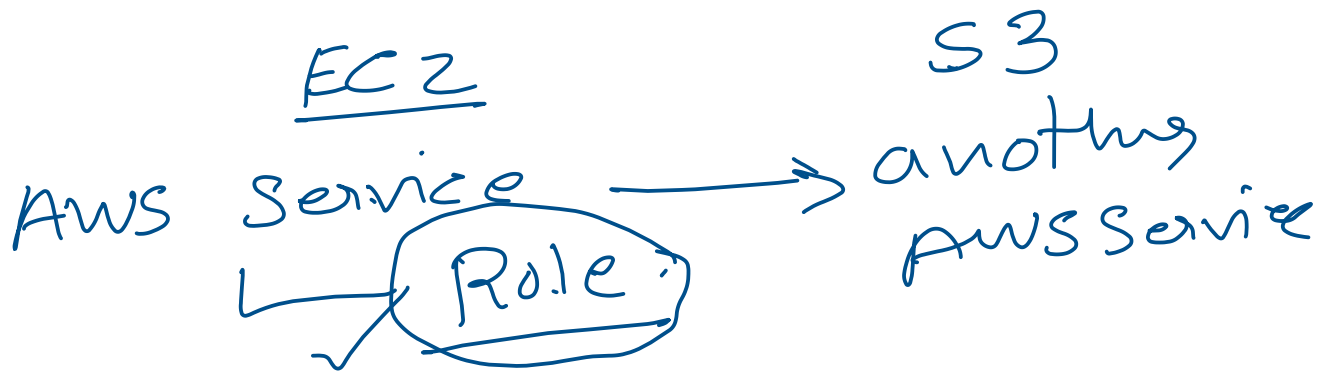
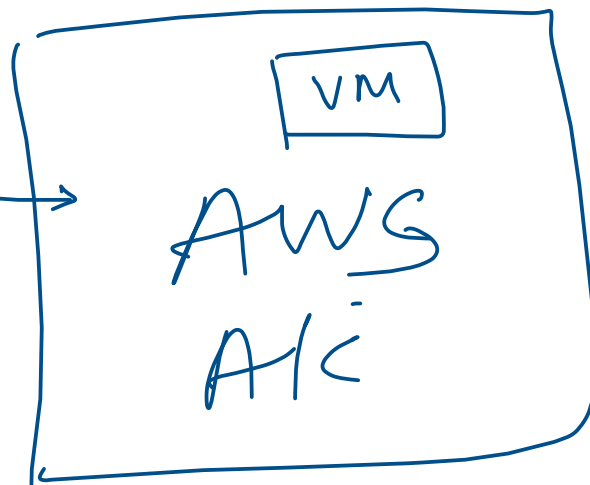
← accessing
Console
Access/Sec

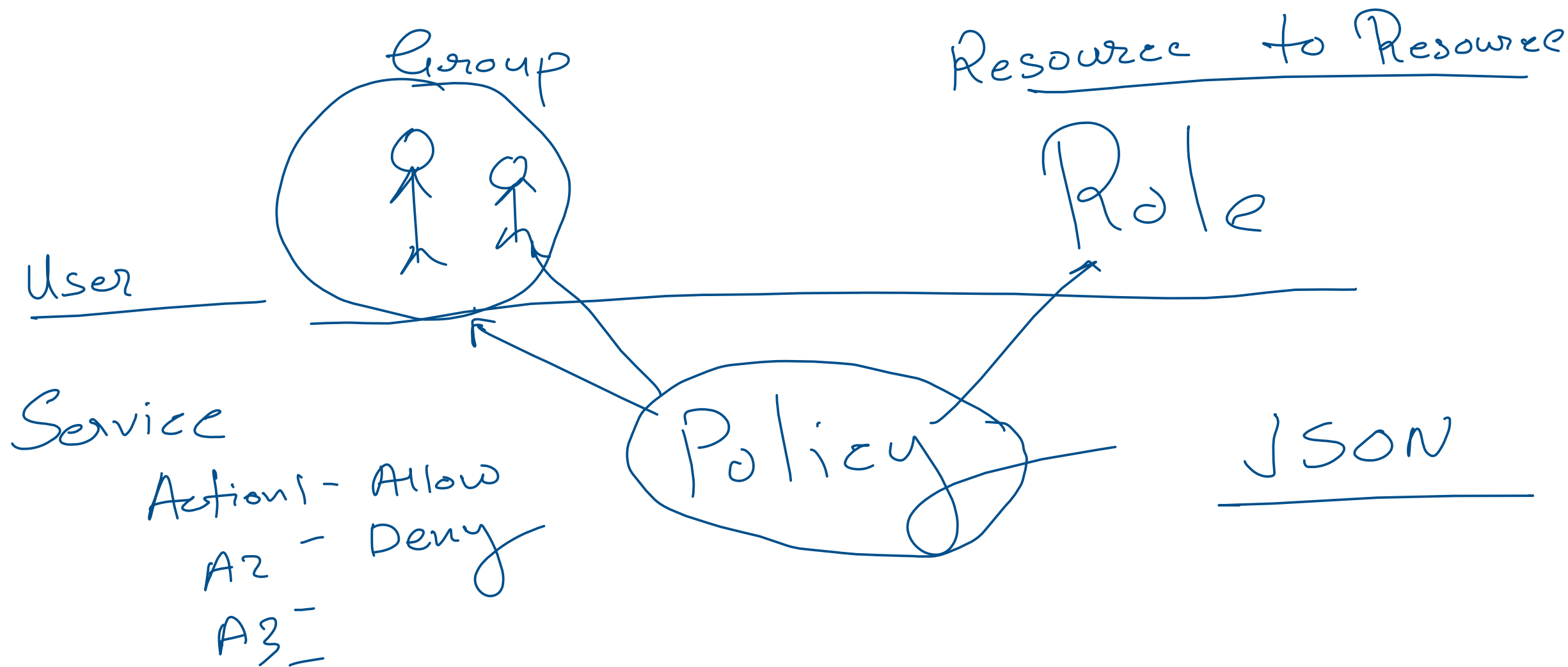
AWS Resources



- java
- c++
python
:
:

encrypt
|
store
Role





EBS — restricted to AZ
1a → EBS
1b.

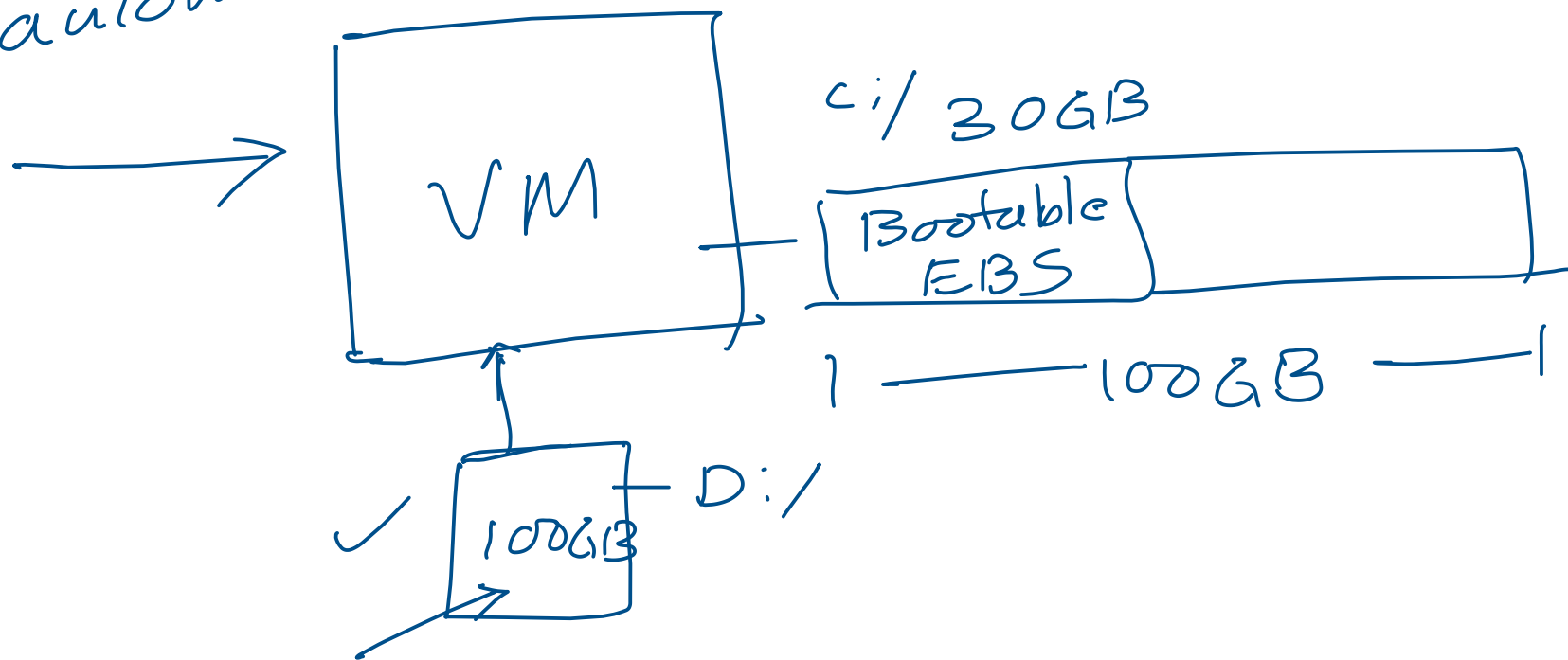
✓ EFS — explicit for Linux (AMI)
— restricted to a Region
— spans across of AZs ↑

✓ FSx — Windows
Regional & across AZs

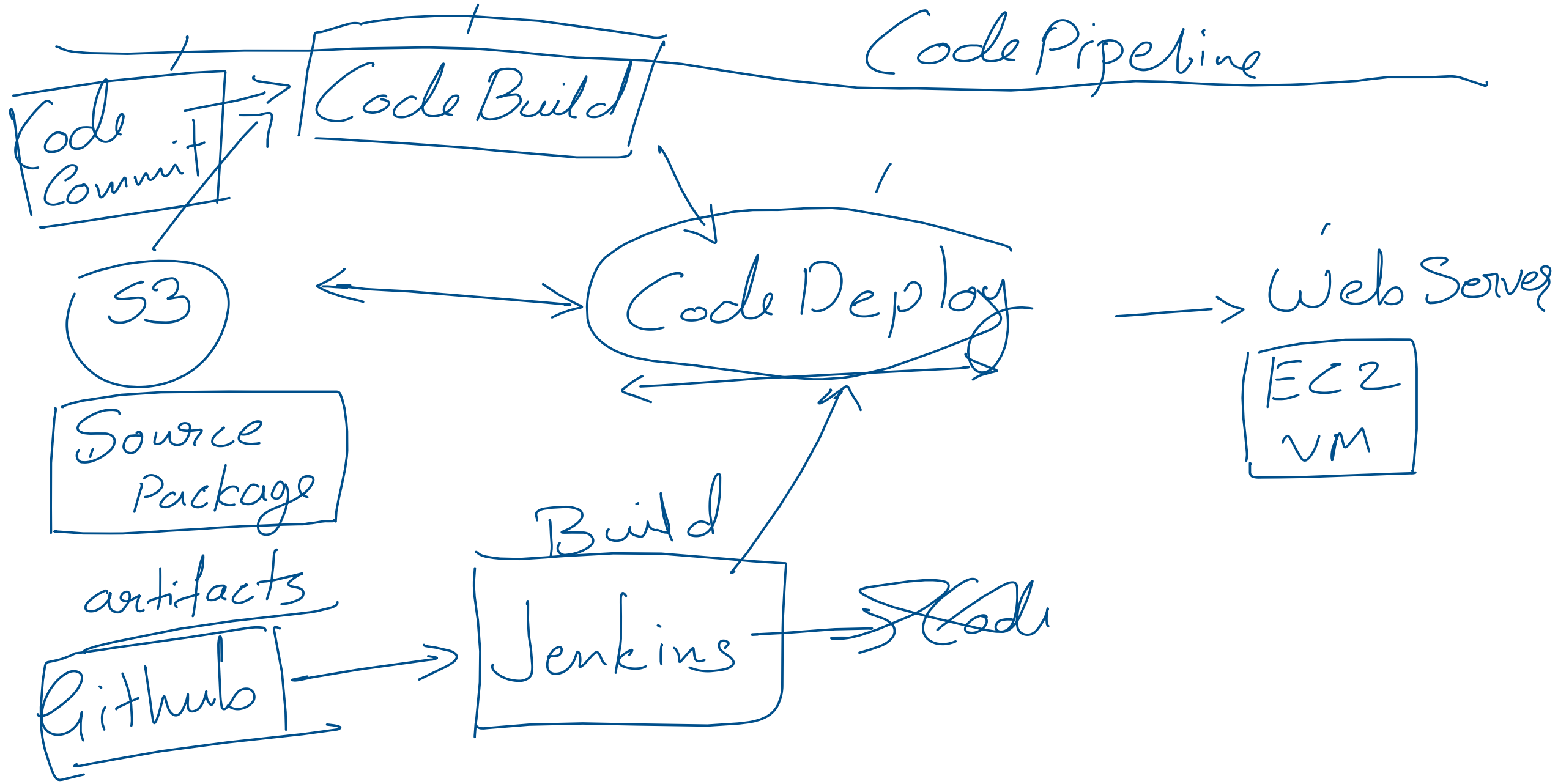
53 — spans across Regions

▽ — access from anywhere

automate



Code Pipeline



DevOps

Engineering

Develops

Operation

Operates

Deploy to
production

vm	rt/w
----	------

Code Deploy

AWS Service

Service Role

